

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

GCSE MATHEMATICS

F

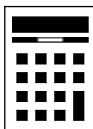
Foundation Tier Paper 3 Calculator

Monday 12 November 2018 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator
- mathematical instruments.



Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

For Examiner's Use

Pages	Mark
2–3	
4–5	
6–7	
8–9	
10–11	
12–13	
14–15	
16–17	
18–19	
20–21	
22–23	
24–25	
26–27	
TOTAL	

Advice

In all calculations, show clearly how you work out your answer.



N 0 V 1 8 8 3 0 0 3 F 0 1

Answer **all** questions in the spaces provided

- 1** Add 8 mm to 7 cm
Circle your answer.

[1 mark]

150 mm

1.5 cm

7.8 cm

708 mm

- 2** In a pie chart, one sector represents $\frac{1}{4}$ of the data.
What is the angle of that sector?
Circle your answer.

[1 mark]

4°

25°

45°

90°

- 3** Which of these **cannot** be the number of lines of symmetry of a triangle?
Circle your answer.

[1 mark]

0

1

2

3



4 Circle the fraction equal to 0.12

[1 mark]

$$\frac{1}{12}$$

$$\frac{3}{25}$$

$$\frac{1}{8}$$

$$\frac{6}{5}$$

5 (a) Solve $n + 7 = 103$

[1 mark]

$$n = \underline{\hspace{4cm}}$$

5 (b) Solve $\frac{m}{6} = 12$

[1 mark]

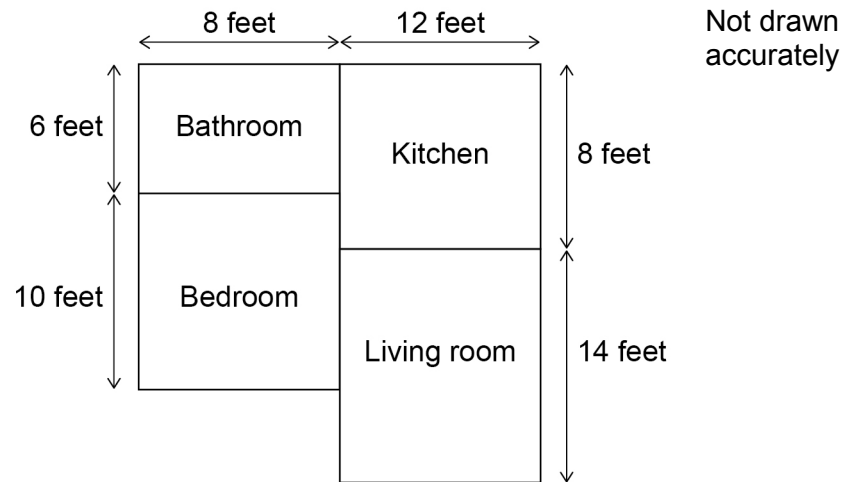
$$m = \underline{\hspace{4cm}}$$

Turn over for the next question



6

Here is a plan of a flat with four rectangular rooms.



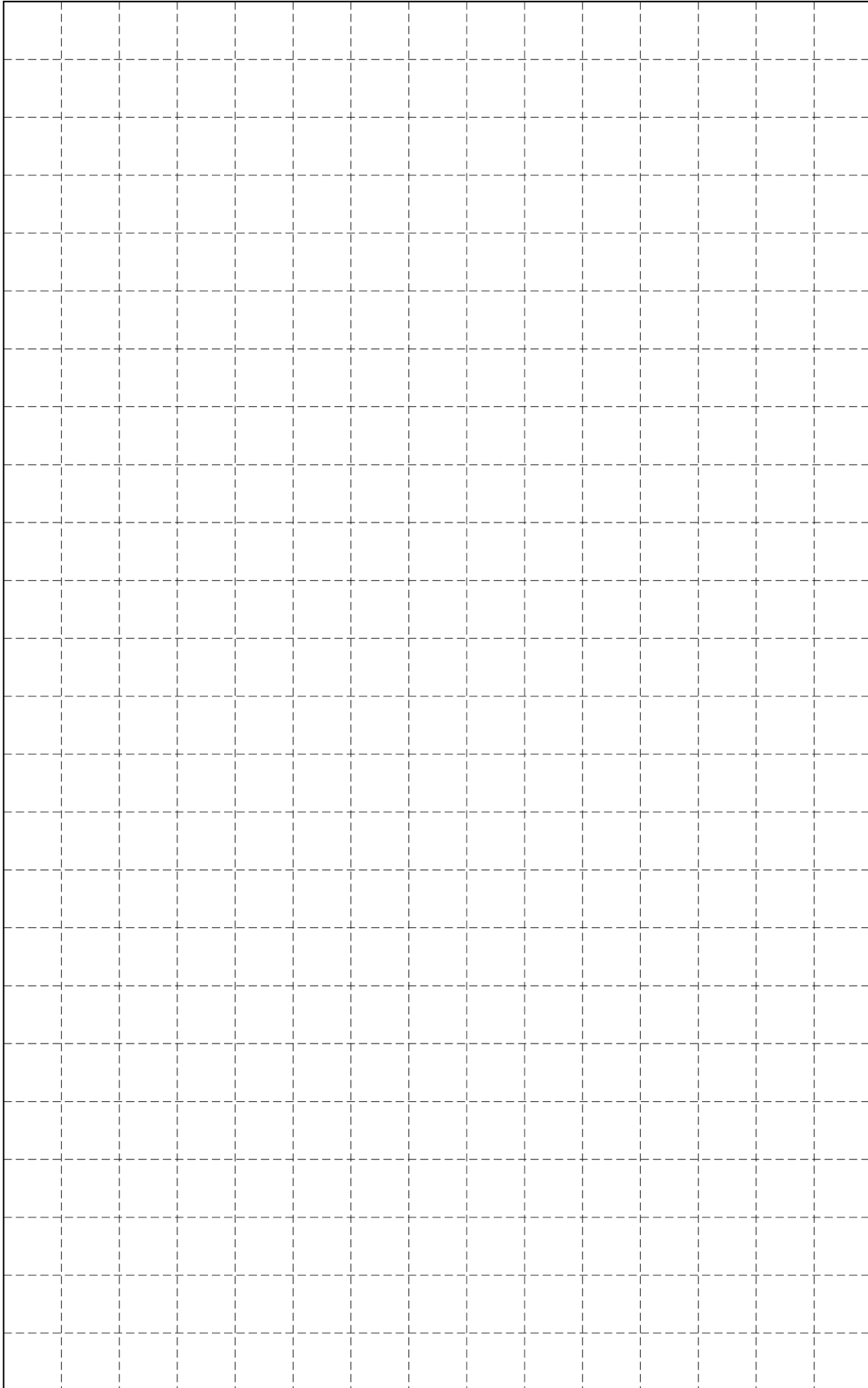
On the grid on the opposite page, make an accurate scale drawing of the plan.

Label each room.

Use a scale of 1 cm represents 2 feet

[3 marks]



Scale: 1 cm represents 2 feet*Do not write
outside the
box***Turn over ►**

7 Here are two groups of numbers, A and B.

Group A

19	11
14	32
16	9

Group B

31	18
28	12

One number is moved from A to B.

The sum of the numbers in B is now 20 **more** than the sum of the numbers in A.

Which number is moved?

You **must** show your working.

[3 marks]

Answer _____



8

Beth sells hot dogs at a market.

Each hot dog is a sausage in a bread roll.



Hot dogs
£3 each

The table shows her costs.

Fee paid to market	£240
Bread rolls	42p per pack of 6
Sausages	£2.50 per jar of 10
Other costs	£57

Beth sells the hot dogs for £3 each.

She sells 300 hot dogs.

Work out her total profit.

[5 marks]

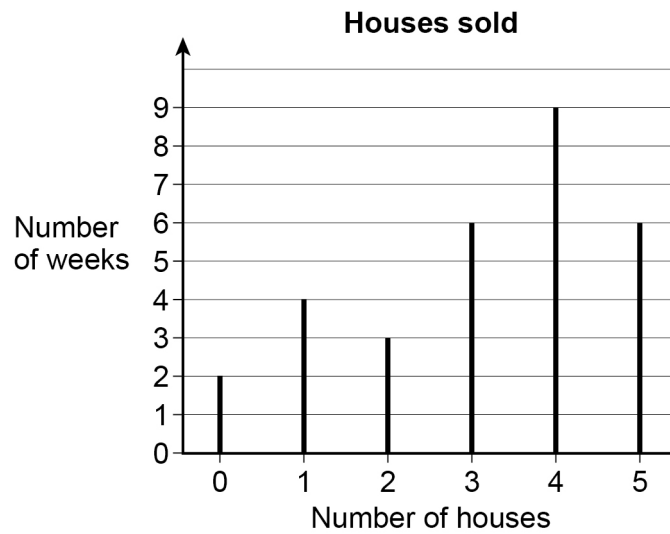
Answer £ _____



9

A company sells houses.

The line graph shows the number sold per week for 30 weeks.



9 (a) Work out the range of the number of houses sold per week.

[2 marks]

Answer _____

9 (b) Work out the median number of houses sold per week.

[2 marks]

Answer _____



9 (c) The company sells three houses.

The prices are £185 000, £239 000 and £136 000

The company earns 2% commission on each house.

In total, how much commission does the company earn on these three houses?

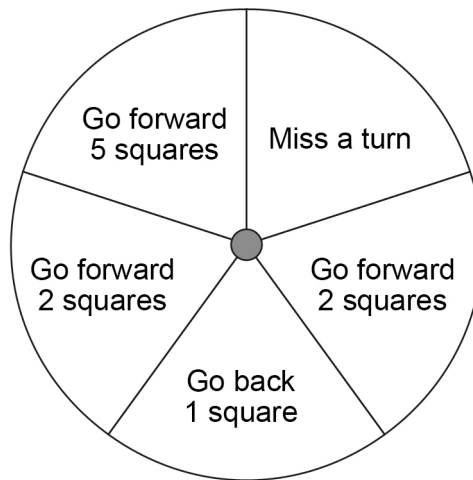
[3 marks]

Answer £ _____

Turn over for the next question



- 10** In a game, a fair spinner has five equal sections as shown.



- 10 (a)** Chloe spins the spinner.

Write down the probability that she gets 'Miss a turn'.

[1 mark]

Answer _____

- 10 (b)** The spinner lands on 'Go back 1 square' three times in a row.
Jamal is next to spin.

Write down the probability that he gets 'Go back 1 square'.

[1 mark]

Answer _____



- 10 (c)** In one game there are 85 spins.

How many of these spins are expected to be 'Go forward 2 squares'?

[2 marks]

Answer _____

- 11** Circle the cube number.

[1 mark]

9

10 000

333

729

- 12** How many minutes is 225 seconds?

Circle your answer.

[1 mark]

$$2\frac{5}{12}$$

$$2\frac{1}{4}$$

$$3\frac{1}{4}$$

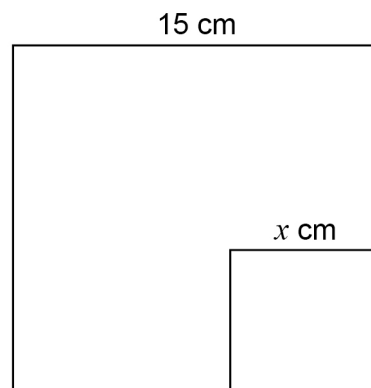
$$3\frac{3}{4}$$



13

A small square has length x cm

A large square has length 15 cm

Not drawn
accurately

The area of the small square is $\frac{1}{9}$ of the area of the large square.

Work out the value of x .**[3 marks]**

Answer _____



- 14 (a)** The term-to-term rule of a sequence is

Add 8 and divide by 2

The first term of the sequence is -24

Work out the next two terms.

[2 marks]

Answer _____ and _____

- 14 (b)** The term-to-term rule of a different sequence is

Subtract 1 and multiply by 5

The third term of this sequence is 120

..... 120

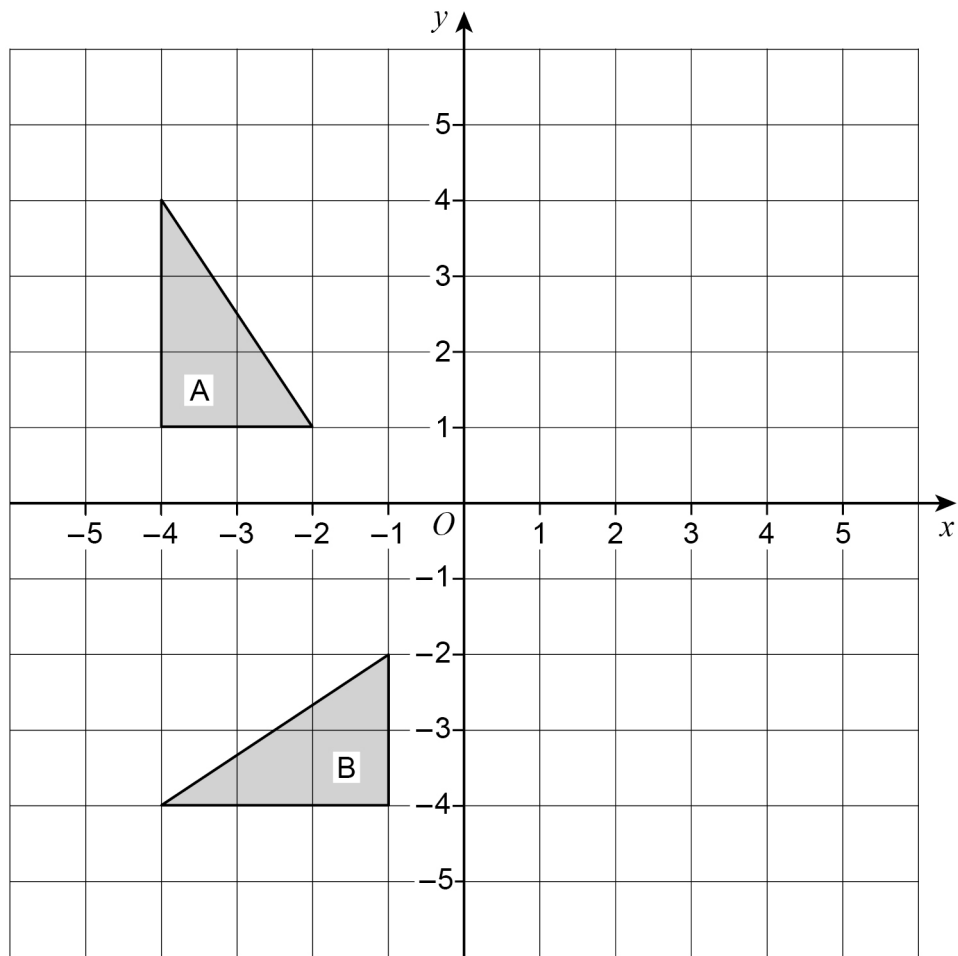
Work out the first term.

[2 marks]

Answer _____



- 15 Describe fully the **single** transformation that maps shape A to shape B. [3 marks]





[4 marks]

[illegible]

Answer £

Turn over for the next question

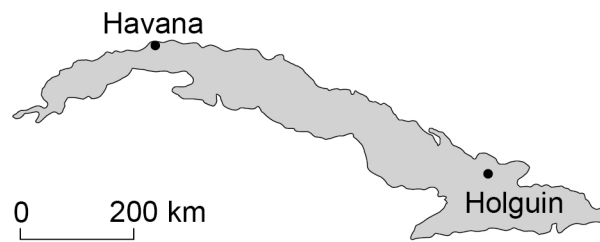
7

Turn over ►



17

Here is a map of Cuba.
1.5 cm represents 200 km



Work out the actual distance from Havana to Holguin.

[3 marks]

Answer _____ km



- 18** Four friends all give each other presents.
The total cost of the presents is £83.40
Work out the mean cost of a present.

[3 marks]

Answer £ _____

Turn over for the next question



- 19** A forest has 6500 trees.
The trees are beech or maple.
number of beech : number of maple = 1.6 : 1

- 19 (a)** What fraction of the trees are beech?

[2 marks]

Answer _____

- 19 (b)** Write number of beech : number of maple in the form $1 : n$

[1 mark]

Answer _____ : _____



20

A shape is translated by the vector $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$

In which direction does the shape move?

Circle your answer.

[1 mark]

up

down

left

right

21

The length of a table is 110 cm to the nearest cm

Complete the error interval.

[2 marks]

_____ cm $\leq$ length < _____ cm

Turn over for the next question

Turn over ►

$$k = n^2 + 9n + 1$$

" k will be a prime number for all integer values of n from 1 to 9"

You **must** show that your value of k is **not** prime.

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

23

At a café,

2 teas and 1 coffee cost £3.40

1 tea and 4 coffees cost £7.30

Work out the cost of 1 tea and the cost of 1 coffee.

[4 marks]

Tea _____

Coffee _____

Turn over for the next question

24

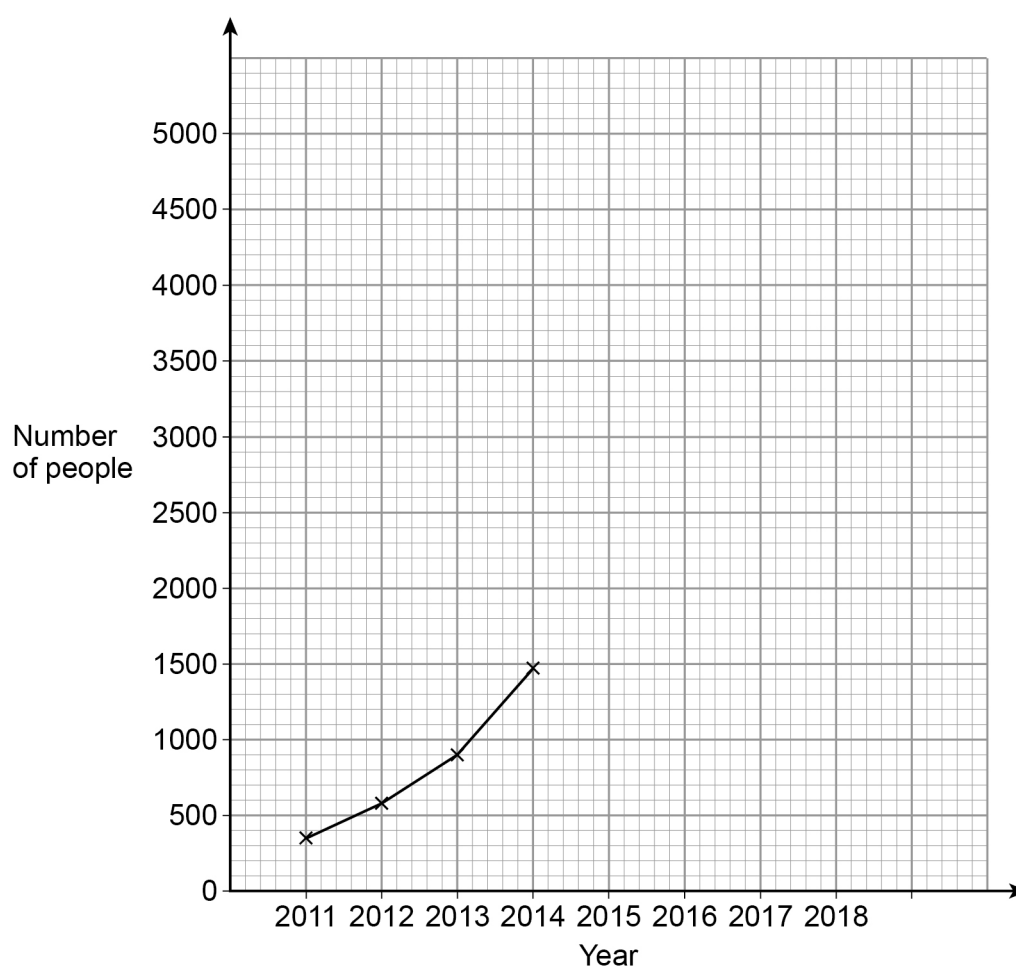
A music festival has taken place each year from 2011

The table shows the number of people who attended each year.

Year	2011	2012	2013	2014	2015	2016	2017	2018
Number of people	350	583	906	1471	2023	2612	3251	3780

The festival organisers draw a time series graph to represent the data.

The first four years have been plotted.



24 (a) Complete the graph.

[2 marks]

24 (b) Use the graph to estimate the number of people who will attend the festival in 2019

[2 marks]

Answer _____

Turn over for the next question



25

Doug owes an amount of £600

He wants to pay back this amount in five months.

He says,

“Each month, I will pay back 20% of the amount I still owe.”

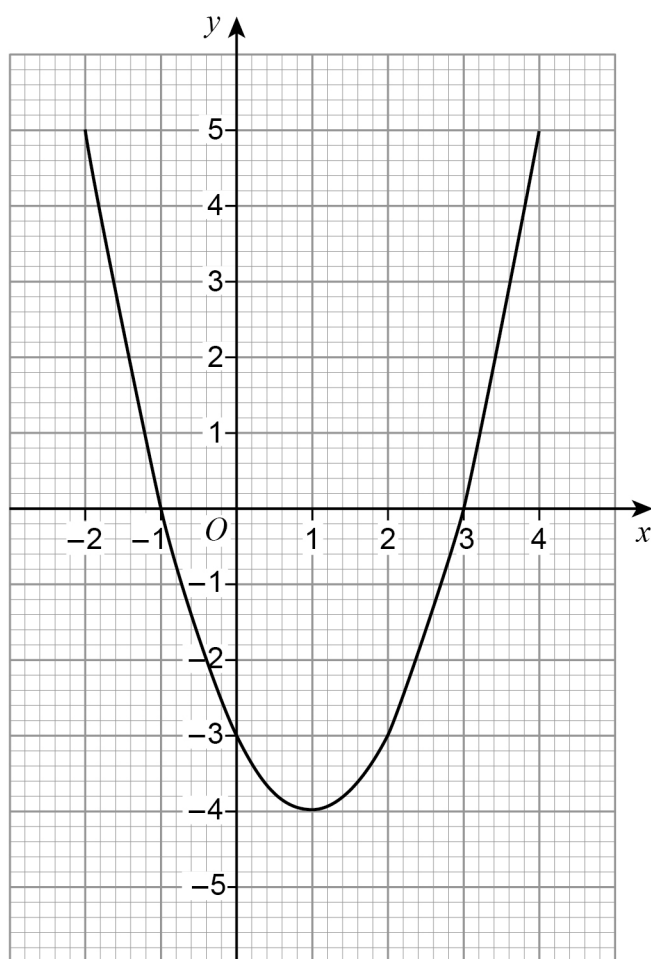
Show working to check if his method is correct.

[3 marks]



26

Here is a quadratic graph.

Circle the x -coordinate of the turning point of the graph.

[1 mark]

-4

-1

1

3

Turn over for the next question

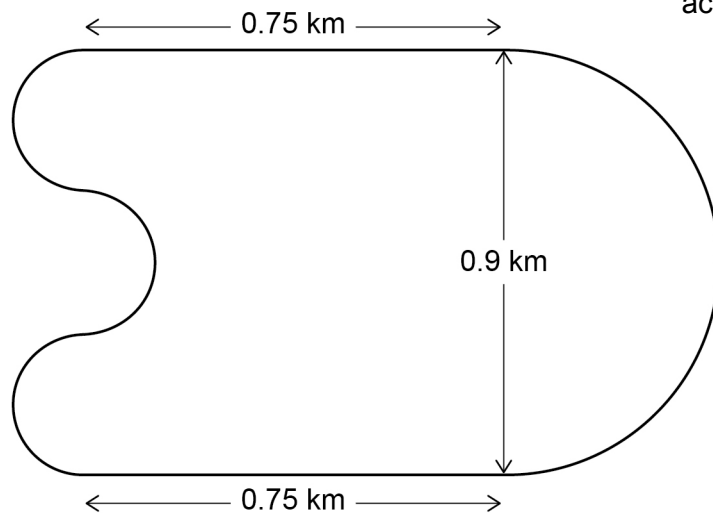
Turn over ►



27

A motor racing circuit consists of
 two parallel straight sections, each of length 0.75 km
 a semicircle of diameter 0.9 km
 three equal, smaller semicircles.

Not drawn
accurately



The length of a motor race must be greater than 305 km

What is the lowest number of **full** laps needed at this circuit?

You **must** show your working.

[5 marks]

Answer _____



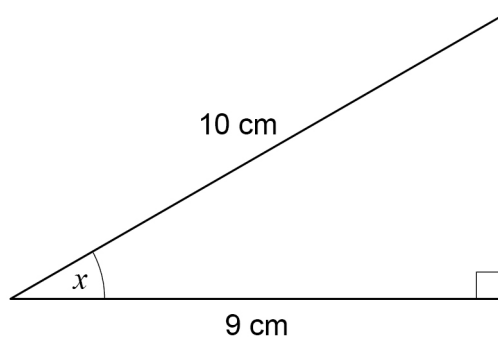
28 Solve $8 > 3 - \frac{1}{2}x$

[2 marks]

Answer _____

29 Use trigonometry to work out the size of angle x .

[2 marks]



Not drawn
accurately

Answer _____ degrees

END OF QUESTIONS

